

Technical Specifications Report: TARUMT Class Replacement System (V2)

1. System Overview & Problem Domain

The TARUMT Class Replacement System is a dedicated web application developed using the Laravel framework and a PostgreSQL relational database. It is engineered to replace a manual, fragmented system based on Google Sheets at the TARUMT Sabah campus. The system specifically addresses operational inefficiencies, multi-cohort scheduling blindspots, data race conditions, and heavy administrative auditing workloads.

The primary objective is to automate the discovery of conflict-free class replacement windows while ensuring strict data and transactional integrity across multiple database records simultaneously.

2. Core Architectural Components

2.1 Multi-Entity Matrix Intersection Engine

The core computational layer of the backend application is the Multi-Entity Matrix Intersection Engine. This engine resolves the problem of mixed student cohorts (such as classes drawing subsets of students from separate programs like RSD3-S1 and RSD3-S2 concurrently) by calculating true availability using relational set intersections.

When a replacement is initiated, the engine executes a deterministic evaluation across four distinct vectors:

- **Lecturer Availability:** The temporal schedule of the requesting academic staff member.
- **Cohort Schedule Intersections:** The combined, overlaid schedules of all linked student groups assigned to that specific module.
- **Physical Venue Tracking:** Global availability tables for physical rooms across the campus property.
- **Capacity-Aware Filtration:** A spatial constraint validator that filters out rooms whose maximum capacity is lower than the total combined student count of the interacting cohorts.

2.2 Optimistic Concurrency Control (OCC) & Smart Conflict Visibility

To safely manage multi-user write operations without relying on performance-heavy pessimistic row locking, the architecture utilizes Optimistic Concurrency Control (OCC). This ensures the web system remains fast and responsive while fully eliminating database double-booking anomalies.

- **Visual Conflict Indicators:** When an available window is rendered on a lecturer's interface, slots that have already been submitted by another faculty member but are still sitting in the administrative queue are explicitly flagged as "Reserved by Another Faculty".

This visual signal guides lecturers away from high-competition windows.

- **Millisecond Validation Check:** The true transaction barrier occurs at the precise millisecond of database submission. When a user submits a booking request, the backend reassesses the slot status inside an isolated database transaction. If a concurrent query successfully changed the slot status to a reservation state even 1 millisecond faster, the slower transaction is instantly aborted, changes are rolled back, and a clear user interface alert is thrown to force a state refresh.

3. User Roles and Functional Specifications Matrix

The system is tailored for a network of Internal Clients and core stakeholders within the university environment. The following matrix outlines the complete distribution of system features across the defined user roles, specifically incorporating user authentication limits and hybrid roles.

User Role	System Access & Functional Capabilities
All Users (Common)	● Secure account authentication (Login and Logout). ● Real-time visualization of personal consolidated master timetables. ● Access to a global replacement update history ledger. ● Automated notification engine receiving immediate alerts for approved schedule shifts.
Students / Student Cohorts	● Authenticated Access: Students possess valid credentials and can log into the system dashboard. ● Strictly View-Only: Authorized functions are entirely restricted to read-only capabilities. Students have no permissions to create, edit, draft, or delete any schedule requests. ● Access to an optimized

User Role	System Access & Functional Capabilities
	acknowledgment view showing finalized cross-cohort synchronized updates and real-time class status tracking.
Lecturers	• Initiation interface to flag a standard timetable block for cancellation or replacement. • Execution control to trigger the multi-entity matrix intersection engine. • Dynamic browsing view displaying available slots alongside "Reserved by Another Faculty" blocks. • Capacity-filtered venue selection matching room sizes to total mixed class numbers. • Draft submission pipeline that triggers immediate OCC checks and assigns a "Pending (You)" state.
Programme Leaders (PLs)	• Administrative Dashboard: Access to a centralized approval interface sorted in a strict First-Come-First-Served (FCFS) order. • One-click digital approval action that updates the slot to "Occupied" and locks the room permanently. • Rejection workflow with a mandatory text reason field that

User Role	System Access & Functional Capabilities
	safely returns the slot to an "Available" state. • Master system configuration panel to handle seed data adjustments for lecturers, cohorts, and classroom caps. • Hybrid Faculty Permissions: Because the 2 Programme Leaders are also active teaching lecturers within the department, they inherit full creation privileges. They can seamlessly Create & Draft Requests for their own course replacements directly inside the interface.

4. Database Design & State Management

The backend database schema utilizes PostgreSQL. For initial validation, verification, and transactional benchmarking during the prototype phase, the relational schema is populated with a highly specialized testing matrix:

- **Lecturer Seeding Constraints:** Exactly 5 unique lecturers are seeded into the database, all belonging strictly to the same department: **Faculty of Computing and Information Technology (FOCS)**.
- **Cohort Seeding Constraints:** 5 separate data-driven student cohorts with predefined group schedules.
- **Asset Allocation:** Standard classroom asset tables with mapped seat limits to govern capacity filters.

The transactional state machine for any given classroom time slot relies on four distinct categorical states mapped within the database tables:

1. **Available:** The slot matches all combined lecturer and student cohort availabilities, and a physical classroom with an adequate seating cap is free.
2. **Pending (Self):** The slot has been drafted and submitted by the active authenticated lecturer and is awaiting administrative execution.
3. **Reserved by Another Faculty:** The slot has been claimed by a different lecturer's

submission query and holds higher chronological priority in the FCFS approval pipeline.

4. **Occupied:** The Programme Leader has officially executed a digital sign-off, or the slot is blocked by the campus master timetable, locking the record entirely from further interactions.